

PART - A

Exp. No.: 1	Introduction to Mitsubishi Industrial Robot
Date:	

Aim:

To acquire knowledge on Mitsubishi industrial robot, its hardware, software, and programming interface.

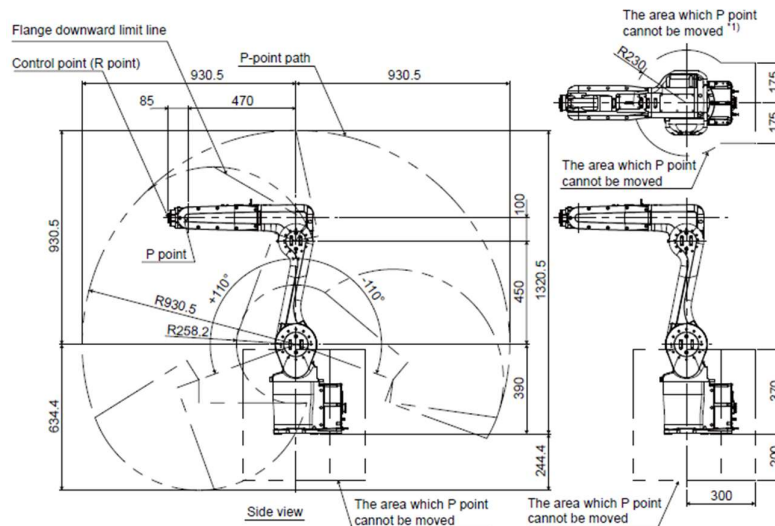
Introduction

• Hardware Specifications and Work envelope

The Mitsubishi Manipulator RV-8CRL is equipped with the necessary hardware and control software to perform precise operations such as pick and place, palletizing, etc. The hardware specifications of Mitsubishi Manipulator RV-8CRL is given in Table 1 and the work envelope is shown in Figure 1.

Type		Unit	RV-8CRL
Environmental specifications			Oil mist
Protection level			IP65
Installation position			On floor, suspended (wall mounted*)
Structure			Vertical articulated
Freedom of motion			6
Drive system			AC servomotor
Position detection system			Absolute encoder
Load capacity	Rating	kg	7
	Maximum	kg	8 ²
Arm length		mm	450+470
Maximum reach radius		mm	931
Installation pitch		mm	□160
Operating range	J1	Degrees	±170
	J2		±110
	J3		+0 to +165
	J4		±200
	J5		±120
	J6		±360
Maximum speed	J1	Degrees/s	288
	J2		321
	J3		360
	J4		337
	J5		450
	J6		720
Maximum composite speed		mm/sec	10,500
Positional repeat accuracy		mm	±0.02
Ambient temperature		°C	0 to 40
Mass		kg	41
Tolerable moment	J4	Nm	16.2
	J5		16.2
	J6		6.86
Tolerable amount of inertia	J4	Kgm ²	0.45
	J5		0.45
	J6		0.1
Tool wiring			15-pin D-SUB
Tool pneumatic pipes			φ6×2
Machine cable			5 m
Connected controller			CR800-D

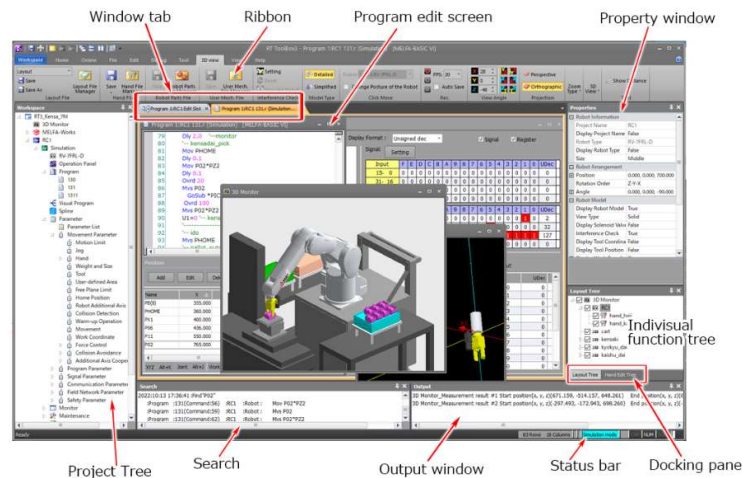
Table 1. Manipulator RV-8CRL specifications



Manipulator RV-8CRL Work envelope

- **Software Tool**

RT ToolBox3 Pro is a powerful robot programming and simulation software developed by Mitsubishi Electric. It supplies a comprehensive suite of tools for creating, editing, debugging, and simulating robot programs to perform the task in the Mitsubishi RV-8CRL manipulator. GUI of RT Toolbox is shown in Figure 2.



RT ToolBox3 operation screen

- **MELFA Basic**

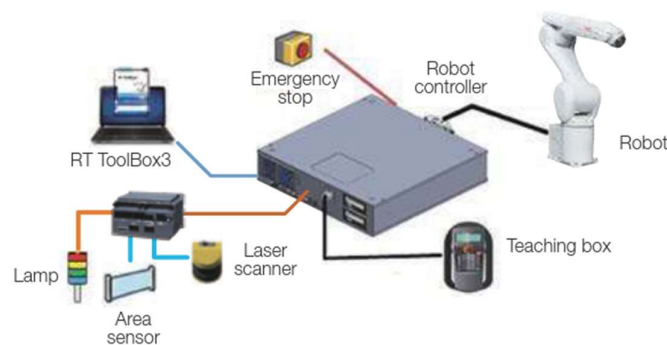
MELFA_BASIC is a high-level programming language designed specifically for programming Mitsubishi MELFA industrial robots. The basic commands of MELFA_BASIC used for Robot programming are given in Table 2.

Table 2. MELFA Basic commands

Serial No	Command	Description
1	MOV	Move the robot from its current position to a specified target position by the shortest route
2	MVS	Move the robot to a specified target position by linear interpolation
3	HOPEN, HCLOSE	Control the opening and closing of the gripper
4	DLY	Introduce a delay or pause in the robot program
5	IF THEN, ELSE, WAIT	For unconditional branching and conditional branching, waiting in the robot program
6	GOTO	Move the program in the indicated line of code
7	M_IN	Input signals
8	END	End of one cycle
9	OVRD	Dynamically adjust the speed and/or acceleration of the robot's motion during program execution

- **Experimental setup & working principle**

The experimental setup displaying the Mitsubishi MELFA industrial robot and its robot controller is depicted in Figure 3. The connection between the PC running RT ToolBox3 and the robot is set up via the CR800 controller. The CR800 series controller functions as the crucial interface and control system for running and managing Mitsubishi industrial robots in a variety of manufacturing applications. Additionally, the teaching box functions as an input device for controlling the robot's movements in Manual mode. In case of an emergency, the emergency stop button is employed to promptly halt the robot's operation.



Experimental Setup

Experimental procedure

Manual Mode:

Performing a simple pick and place task on the Industrial robot using the teach pendant or programming interface to guide the robot through the desired motions. Here is a general outline of the experiment procedure:

- Ensure that the robot and its controller are powered on and in a ready state.
- Ensure that Manual mode is selected in Teach pendant and Controller.
- Initialize the robot to a safe and known state. This typically involves moving the robot to a predefined home or starting position.
- Use the teach pendant to manually guide the robot through the pick and place waypoints. The waypoints include positions where the robot should pick up and place objects.
- At each waypoint, record the position and orientation information.
- Refine the robot's movements by adjusting the speed and trajectory during the teaching process. Ensure that the robot moves smoothly and accurately to each waypoint.
- Verify that the gripper operations (open/close) are functioning as intended during the pick and place task.
- Familiarize yourself with the emergency stop procedures on the teach pendant. Ensure that you know how to stop the robot in case of unexpected situations.
- Repeat the pick and place task several times to validate the taught waypoints and ensure consistent performance.
- Fine-tune the robot's movements and gripper actions based on your observations during the manual operation.
- Save the recorded waypoints and gripper commands for future use. These will be used as a basis for creating a program for the autonomous execution of the pick and place task.
- Exit the manual mode on the teach pendant when the teaching process is complete.

Auto Mode:

Performing a simple pick and place task in automatic mode on the Industrial robot involves creating a program that defines the sequence of movements for the robot to execute the task autonomously. Here's a general outline of the experiment procedure:

- Ensure that the Industrial robot and its controller are powered on and in a ready state.
- Ensure that Auto mode is selected in Teach pendant and Controller.
- Access the programming interface or teach pendant of the robot.
- Start a new program within the programming environment. This program will have instructions for the pick and place task by defining waypoints, set the parameters and gripper action.
- Execute the program in simulation mode or with the robot in a controlled environment to verify that the movements and gripper actions are as expected.
- Refine the program iteratively, adjusting to improve the efficiency and accuracy of the pick and place task.

- Implement safety measures within the program, such as emergency stop routines and safety checks, to ensure safe operation during autonomous execution.
- Once satisfied with the program, load it onto the robot controller.
- Initiate the program execution on the robot and see its performance as it autonomously carries out the pick and place task.
- Monitor the robot during execution and be prepared to troubleshoot any issues that may arise.

RT Toolbox3 Program:

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Results and Discussion: